

```
In from io import IncrementalDownloadDecoder
[204]: from sentence_transformers import SentenceTransformer
from PIL import Image
import matplotlib.pyplot as plt
import numpy as np
from plotly import offline
model = SentenceTransformer("clip-ViT-B-32")

Loading weights: 0% | 0/398 [00:00<?, 111/1]
```

```
In img1 = "/content/5.jpeg"
[205]: img1 = Image.open(img1)
plt.figure(figsize = (6,6))
plt.imshow(img1)
plt.title("Image For Similarity Comparison",fontsize=14)
plt.show()
```



```
In text_description = ["Two dogs in the snow",
[206]: "A cat on a table",
"A Picture of London at night",
"A dog sleeping on a couch",
"Two cats fighting in the garden",
"This image features Black Panther, the Marvel superhero, in a dramatic watercolor splash-art style",
"The image is a close-up side profile of a futuristic humanoid robot with a sleek white outer shell",
"The robot has a human-like face with smooth features and a calm expression",
"This image shows a striking black wolf looking directly at the viewer",
"The peacock stands proudly, often associated with royalty and divine elegance",
"Known as the "Palace of Winds," this Jaipur landmark features a honeycomb facade of pink and red sandstone",
"A serene marble structure dedicated to Lord Krishna, adorned with ornate carvings and domes",
"Its massive walls, watchtowers, and gateways tell stories of valor and royal legacy, overlooking arid landscapes and ancient towns.",
"The peacock combines the elegance of a peacock, the grandeur of the Hawa Mahal",
"Each element is placed in harmony, creating a vivid snapshot of India's heritage."

]
```

```
In def display_image_and_compare(image_path,text_description):
[207]: # Loading Image
img = Image.open(image_path)

# Encode Text and Image (Converting both into Numbers)
text_emb = model.encode(text_description)
img_emb = model.encode(img)

# Compute Similarity Score
similarity_score = np.dot(text_emb,img_emb.T)

# Find the Best Match
best_match_index = np.argmax(similarity_score)
best_match_desc = text_description[best_match_index]

# Print the final Output
print(f'Best match for the image is = {best_match_desc} with score of = {similarity_score.max()}')

# Display the Image
plt.figure(figsize = (6,6))
plt.imshow(img1)
plt.axis('off')
plt.title("Image For Similarity Comparison",fontsize=14)
plt.show()
```

```
In display_image_and_compare("/content/5.jpeg",text_description)
[208]: Best match for the image is = This image features Black Panther, the Marvel superhero, in a dramatic watercolor splash-art style = with score of = 26.91294343668945
```



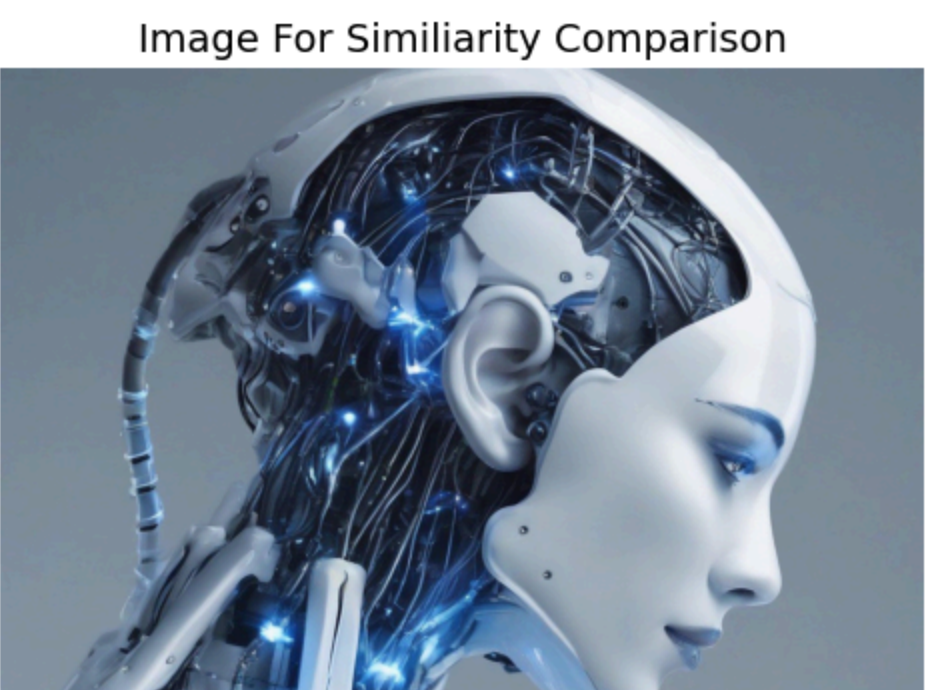
```
In img1 = "/content/3.jpeg"
[221]: img1 = Image.open(img1)
plt.figure(figsize = (6,6))
plt.imshow(img1)
plt.axis('off')
plt.title("Image For Similarity Comparison",fontsize=14)
plt.show()
```



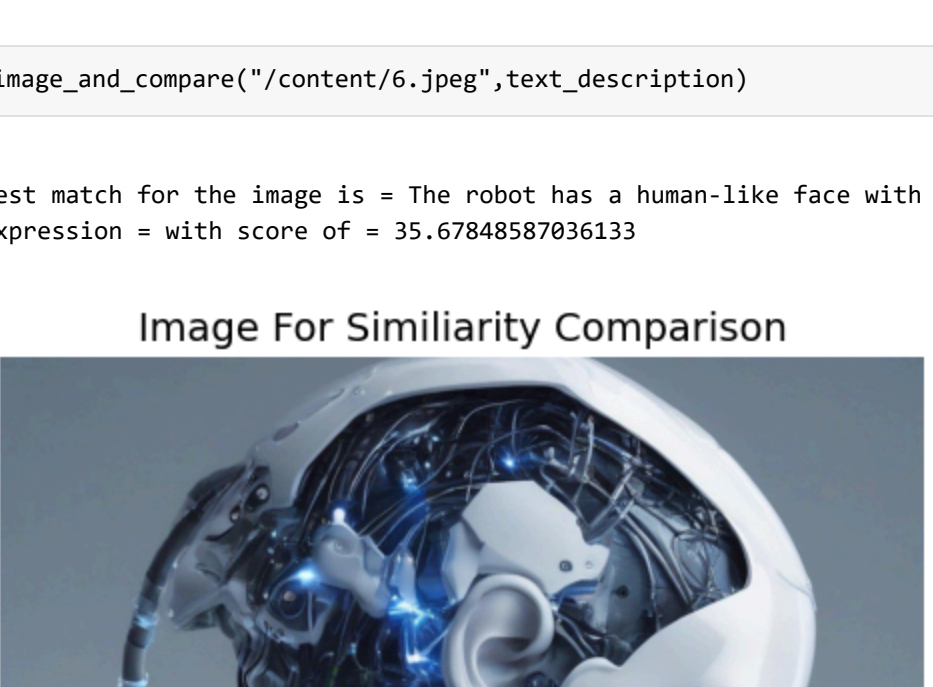
```
In display_image_and_compare("/content/3.jpeg",text_description)
[208]: Best match for the image is = A Cat on a table = with score of = 28.897214889526367
```



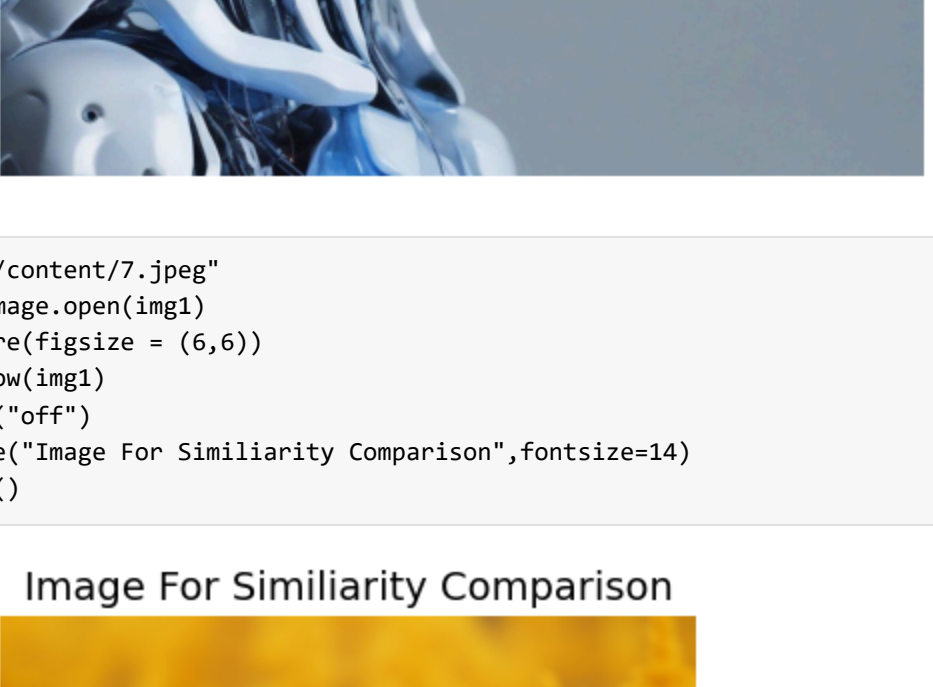
```
In img1 = "/content/6.jpeg"
[217]: img1 = Image.open(img1)
plt.figure(figsize = (6,6))
plt.imshow(img1)
plt.axis('off')
plt.title("Image For Similarity Comparison",fontsize=14)
plt.show()
```



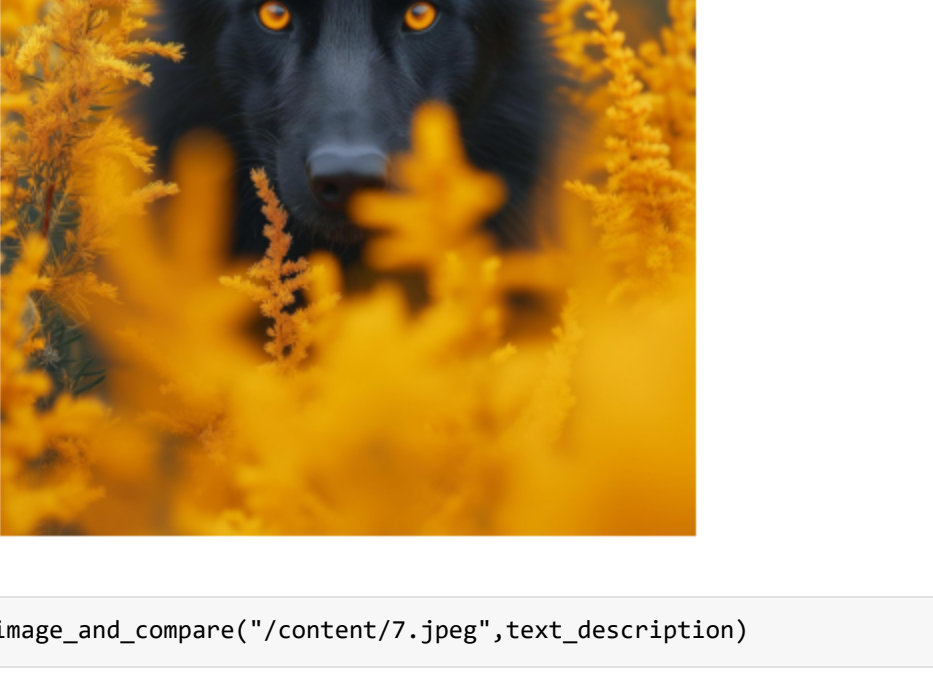
```
In display_image_and_compare("/content/6.jpeg",text_description)
[208]: Best match for the image is = The robot has a human-like face with smooth features and a calm expression = with score of = 35.67849387936133
```



```
In img1 = "/content/7.jpeg"
[205]: img1 = Image.open(img1)
plt.figure(figsize = (6,6))
plt.imshow(img1)
plt.axis('off')
plt.title("Image For Similarity Comparison",fontsize=14)
plt.show()
```



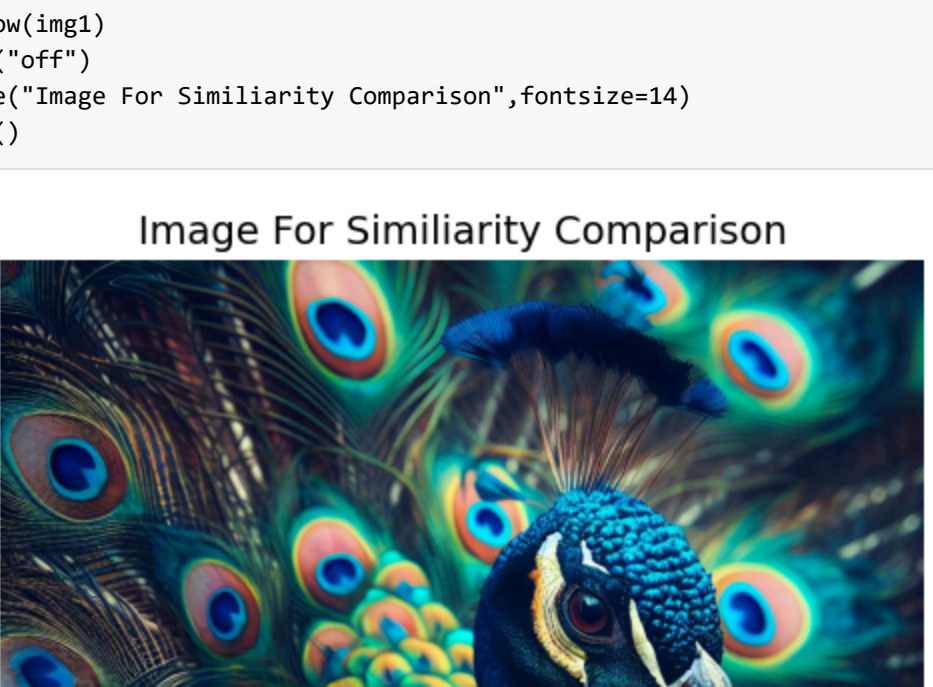
```
In display_image_and_compare("/content/7.jpeg",text_description)
[206]: Best match for the image is = This image shows a striking black wolf looking directly at the viewer = with score of = 29.462349113201895
```



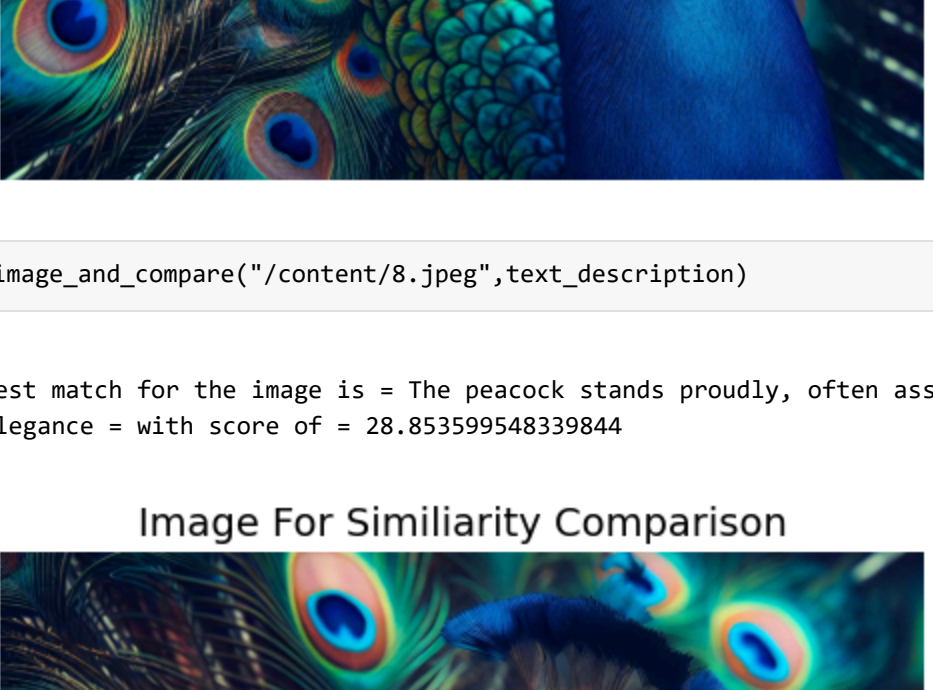
```
In img1 = "/content/8.jpeg"
[207]: img1 = Image.open(img1)
plt.figure(figsize = (6,6))
plt.imshow(img1)
plt.axis('off')
plt.title("Image For Similarity Comparison",fontsize=14)
plt.show()
```



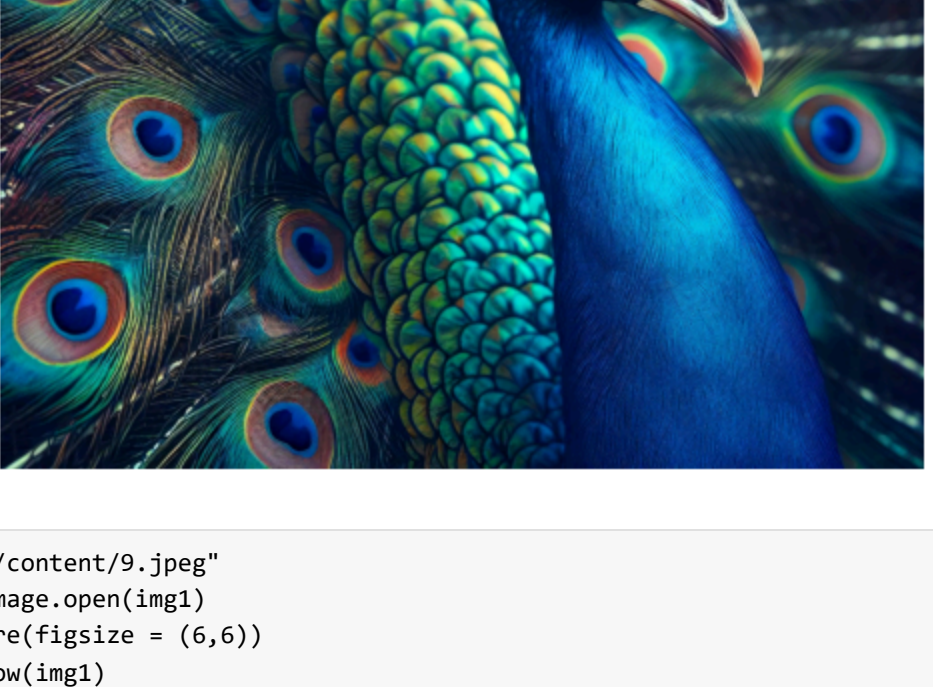
```
In display_image_and_compare("/content/8.jpeg",text_description)
[208]: Best match for the image is = The peacock stands proudly, often associated with royalty and divine elegance = with score of = 28.83309548339844
```



```
In img1 = "/content/9.jpeg"
[209]: img1 = Image.open(img1)
plt.figure(figsize = (6,6))
plt.imshow(img1)
plt.axis('off')
plt.title("Image For Similarity Comparison",fontsize=14)
plt.show()
```



```
In display_image_and_compare("/content/9.jpeg",text_description)
[210]: Best match for the image is = Its massive walls, watchtowers, and gateways tell stories of valor and royal legacy, overlooking arid landscapes and ancient towns = with score of = 22.705873962480314
```



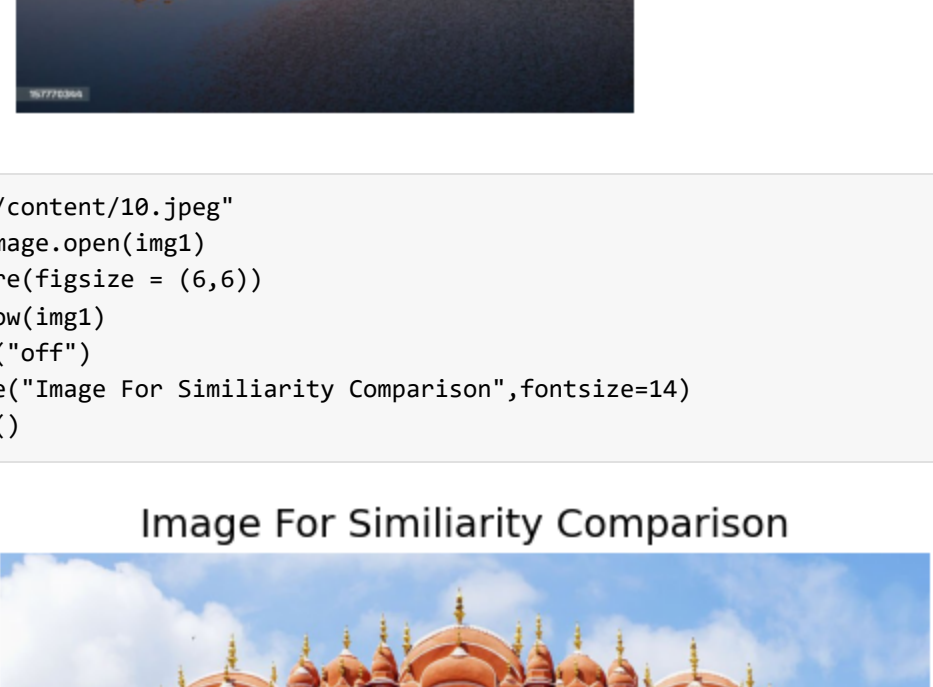
```
In img1 = "/content/10.jpeg"
[211]: img1 = Image.open(img1)
plt.figure(figsize = (6,6))
plt.imshow(img1)
plt.axis('off')
plt.title("Image For Similarity Comparison",fontsize=14)
plt.show()
```



```
In display_image_and_compare("/content/10.jpeg",text_description)
[210]: Best match for the image is = Each element is placed in harmony, creating a vivid snapshot of India's heritage = with score of = 25.2757568359375
```



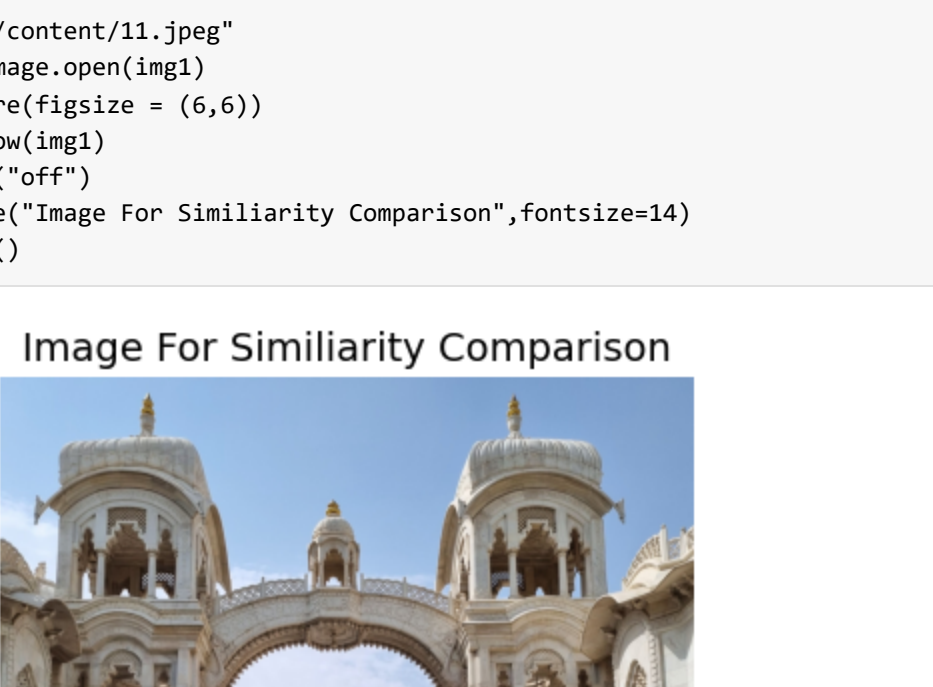
```
In img1 = "/content/11.jpeg"
[213]: img1 = Image.open(img1)
plt.figure(figsize = (6,6))
plt.imshow(img1)
plt.axis('off')
plt.title("Image For Similarity Comparison",fontsize=14)
plt.show()
```



```
In display_image_and_compare("/content/11.jpeg",text_description)
[214]: Best match for the image is = Each element is placed in harmony, creating a vivid snapshot of India's heritage = with score of = 27.427565282820112
```



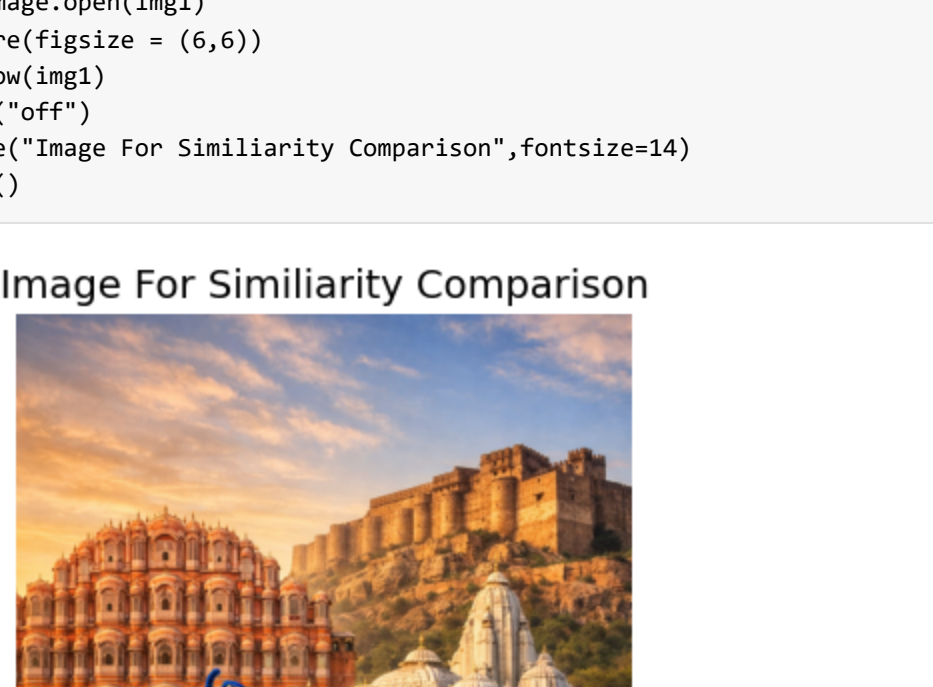
```
In img1 = "/content/12.jpeg"
[215]: img1 = Image.open(img1)
plt.figure(figsize = (6,6))
plt.imshow(img1)
plt.axis('off')
plt.title("Image For Similarity Comparison",fontsize=14)
plt.show()
```



```
In display_image_and_compare("/content/12.jpeg",text_description)
[216]: Best match for the image is = Each element is placed in harmony, creating a vivid snapshot of India's heritage = with score of = 29.319156646728916
```



```
In display_image_and_compare("/content/13.jpeg",text_description)
[216]: Best match for the image is = Each element is placed in harmony, creating a vivid snapshot of India's heritage = with score of = 29.319156646728916
```



```
In display_image_and_compare("/content/13.jpeg",text_description)
[216]: Best match for the image is = Each element is placed in harmony, creating a vivid snapshot of India's heritage = with score of = 29.319156646728916
```

